MCP: one way to plug tools in

The Model Context Protocol, so every tool doesn’t need its own integration with every AI app.

60 min | Recorded | Class 11 of 15

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What this session covers

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2 slides

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What a server offers
2 slides

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Using it safely
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4 slides

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Section 1 of 4

The problem MCP fixes

Every app times every tool.

01

Without a standard, every AI app needs its own connector to every system

02

Three parts, like a plug and a socket

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Without a standard, every AI app needs its own connector to every system

Before

- 3 AI apps × 5 of our systems = 15 integrations
- Each built differently, each maintained separately
- A Tracklane change breaks all of them

With MCP

- Each system gets one MCP server
- Each app speaks MCP once
- 3 + 5 = 8 pieces, and they all fit

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Three parts, like a plug and a socket

The host is the app people use; the server is what we build.

01 Host
The AI app: a chat app, a code editor, our own agent. It runs the model.

02 Client
Lives inside the host. One per server it connects to.

03 Server
Ours. Exposes Tracklane lookups, runbooks, prompts. Knows nothing about which model is asking.

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Section 2 of 4

What a server offers

Tools, resources and prompts.

01

A server can offer three kinds of thing

02

Two ways to connect: on the same machine, or over the network

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A server can offer three kinds of thing

Tools	Actions the model can ask for: look up a shipment, check a batch. Same idea as class 10.
Resources	Things to read: a runbook, a document, a record. The host decides when to include them.
Prompts	Ready-made instructions a person picks from a menu: “triage this ticket”.

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Two ways to connect: on the same machine, or over the network

Local (stdio)
The host starts the server as a program on your laptop. Simple, private, one user.

Remote (HTTP)
The server runs somewhere central. Shared by a team, needs sign-in and access rules.

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Section 3 of 4

Using it safely

A server is code you run.

01

MCP servers go through the same rules as any other tool

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A server is **code you run**, so treat it that way.

Installing an MCP server is installing software. It can read what it’s allowed to, and do what it’s allowed to.

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MCP servers go through the same rules as any other tool

- Only servers in our directory, each with a data ceiling (class 8)
- Read what a server’s tools actually do before connecting it
- Remote servers sign users in; they don’t share one key
- Watch for tool descriptions that try to give the model orders

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Section 4 of 4

Code walkthrough

A small server for operations.

01

One tool, one resource and one prompt in about twenty lines

02

Adding it to a host is a few lines of config

03

Where today sits on the card

04

What to do next

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One tool, one resource and one prompt in about twenty lines

code/class11/server.py

```python
mcp = FastMCP("lol-ops")

@mcp.tool()
def get_shipment(shipment_id: str) -> dict:
    """Where a LOL shipment is and why, e.g. LOL-SH-4472190."""
    return SHIPMENTS.get(shipment_id, {"error": "no such shipment"})

@mcp.resource("runbook://{name}")
def runbook(name: str) -> str:
    return (RUNBOOKS / f"{name}.md").read_text(encoding="utf-8")

@mcp.prompt()
def triage(ticket: str) -> str:
    return f"Route this LOL ticket to one queue and say why in one sentence:\n\n{ticket}"
```


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Adding it to a host is a few lines of config

code/class11/host-config.json

```
{
  "mcpServers": {
    "lol-ops": { "command": "uv", "args": ["run", "code/class11/server.py"] }
  }
}
```


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Where today sits on the card

1. A script	The steps are known and don’t change.
2. A rule	The decision is known and must be explainable line by line.
3. Classical ML	Labelled history, and the input is a table or short text.
4. A small trained model	Lots of text, images or audio, enough labels, cost per call matters.
5. An LLM with a prompt	Varied language in and out, few labels, someone checks the output.
6. An LLM plus retrieval, tools or agents	**Needs private knowledge, has to act, or takes several steps.**
7. Not yet	No ground truth, needs a guarantee, or nobody owns the result.

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What to do next

1. Connect `server.py` to an MCP host you're allowed to use and ask about LOL-SH-4472190.
2. Add a resource for the handwritten tickets, then decide whether it should exist.
3. Read the class 12 pre-reading on agents.

Questions after class: the course channel.

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